

Uncertainty-aware haplotype phasing of genetic and epigenetic variation from long reads

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Abstract

Long-read phasers rarely combine sequence variants with the methylation signal on the same reads, and seldom report which phase assignments are unreliable. Here we present LongPhase 2, which phases SNVs, indels, structural variants and 5-methylcytosine in one graph, weights observations by base quality and sequence context, and scores every variant's phasing uncertainty. A graph neural network identifies phase-disagreement onsets for selective unphasing. Scored against a benchmark from the telomere-to-telomere HG002 assembly, LongPhase 2 makes 2.8- to 4.8-fold fewer switch errors than WhatsHap and HapCUT2 on nanopore data while running five- to sixfold faster on one thread at 60×, and 2.2- to 2.6-fold fewer than WhatsHap on PacBio HiFi data without retraining. Co-phasing all four classes increases block N50 at 60× by 43% relative to SNV-only phasing. Against the earlier, mapping-based benchmark, the tools differ only 1.2- to 1.3-fold at high coverage, and most of their switch errors coincide. LongPhase 2 keeps the command-line interface and output formats of LongPhase, which Clair3, ClairS and EPI2ME call.

Introduction

Haplotype phasing, the assignment of heterozygous variants to their parental chromosomes, is essential for interpreting the diploid human genome. Whether two variants act in *cis* or in *trans* decides compound heterozygosity in recessive disease¹, and allele-specific methylation on a phased haplotype reveals its parent of origin². Long-read sequencing has made read-based phasing practical at clinical and population scale. Individual reads span tens to hundreds of kilobases, link distant heterozygous sites directly and carry native base-modification signals; a single library thus yields haplotype blocks orders of magnitude longer than short reads allow^{3,4}. Read-based phasing does not require parental sequencing, a population reference panel of the kind statistical phasing depends on⁵, or the additional Hi-C or Strand-seq libraries used to phase whole chromosomes⁶.

Read-based phasers optimize agreement between reads and candidate haplotypes through weighted minimum error correction or probabilistic models, as in WhatsHap⁷, HapCUT2⁸ and Margin⁹. These methods centre on single-nucleotide variants (SNVs), although WhatsHap, by re-aligning each read to both alleles, also co-phases small and large indels with them¹⁰. We previously introduced LongPhase¹¹, which recasts phasing as a weighted allele graph, resolves

haplotypes at chromosome scale and co-phases SNVs with structural variants (SVs)^{12,13}. WhatsHap and LongPhase are used in several long-read analysis pipelines (Supplementary Table 6). Clair3 phases with WhatsHap by default and with LongPhase as an option, and the somatic callers ClairS and ClairS-TO use LongPhase by default^{14–17}; the EPI2ME wf-human-variation workflow phases every nanopore genome with LongPhase¹⁸; and the SV callers Sniffles2 and Severus take as input the haplotagged alignments that either phaser writes^{12,19}. Phasing errors can therefore propagate, unflagged, to downstream analyses.

Three challenges remain in read-based phasing. The first is integrating 5-methylcytosine (5mC), which long reads report alongside the sequence, into haplotype inference. Where the two haplotypes differ in methylation, as at imprinted loci and at sites of sequence-dependent allele-specific methylation, a read’s CpG states tie it to one haplotype even where it spans no heterozygous variant, which is where sequence-based blocks end^{20,21}. Several methylation-aware tools use that signal once a sequence phase exists. NanoMethPhase assigns methylation to haplotypes that WhatsHap has already resolved²⁰; MethPhaser and HapBridge learn haplotype-specific CpGs from the reads an SNV phaser has tagged, then join adjacent blocks or flag suspected switches^{21,22}; and LongHap, which co-phases SNVs, indels and SVs, adds methylation only to bridge the gaps its sequence phasing leaves²³. Because the informative CpGs are defined by the preceding sequence phase, its errors propagate into the methylation evidence, and methylation can revise a phase only after it is built. Allele-specific methylation is moreover confined to a small fraction of CpGs, varies between cells and tissues, and is read as a per-molecule probability rather than a base^{20,21}; joint modelling allows such sparse, noisy evidence to be weighed together with the sequence alleles on the same reads.

The second challenge is accounting for errors that depend on genomic context. WhatsHap weights observations in its minimum-error-correction objective, and HapCUT2 incorporates base-quality information in its likelihood model^{7,8}. Base quality alone, however, cannot establish that a read originates from the locus to which it is aligned. Base-calling errors concentrate in homopolymers and tandem repeats, reads misplaced in segmental duplications create spurious linkages, and heterozygous calls in copy-number-altered regions can mislead the phasing graph^{24,25}. These context-dependent errors can create switches within haplotype blocks and confound compound-heterozygosity calls and haplotype-aware somatic detection^{16,19}.

A third challenge is identifying unreliable phase assignments. Statistical phasers attach posterior probabilities to phased genotypes⁵, and HapCUT2 uses posterior confidence to prune variants and split low-confidence blocks⁸. Such confidence estimates depend on the phasing model and may remain high when misplaced reads provide internally consistent evidence. Local sequence context and the surrounding evidence therefore provide additional information for assessing phase reliability. WhatsHap and Margin report no confidence, and LongPhase reports only simple read support, the number of a read’s alleles on each haplotype and a Phred-scaled tagging quality derived from them, not the reliability of each variant’s phase. Learning-based phasers, built on graph neural networks or deep reinforcement learning^{26,27}, also provide no per-variant reliability estimate. They learn, partly from simulated reads, to optimize the same fragment-cut or minimum-error-correction objective as heuristic phasers and phase SNVs only, and their authors note that errors from mis-alignment in repetitive regions remain unresolved²⁷.

Phasers are commonly benchmarked against Genome in a Bottle (GIAB) truth sets built largely by mapping reads to GRCh38^{28,29}, which assess mainly the regions where mapping-based methods perform well³⁰. The HG002 v4.2.1 release omits both sex chromosomes and 12% of the autosomes, mostly satellite arrays and segmental duplications^{29,30}, where mis-mapped reads and collapsed paralogues create false linkages^{24,25}. Its phase is also partial: the original release recorded phase only for the 74% of heterozygous variants on which trio-based phasing agreed with Strand-seq and HiFi phasing by WhatsHap²⁹, so the variants hardest for a read-based phaser are the least likely to be scored (the phase-transferred version scored here is described in Methods). Extending benchmarks into difficult sequence has changed conclusions before. GIAB’s growth from 85% to 92% of GRCh38 exposed eight times more false negatives in a short-read call set²⁹, and the complete CHM13 reference removed tens of thousands of spurious variant calls per sample³¹. The complete diploid T2T-HG002 assembly extends the available phase information³⁰. It resolves both haplotypes of every chromosome, adds 701 Mb of autosomal sequence absent from v4.2.1 and phases every heterozygous site. Its projection onto GRCh38, the GIAB v5.0q benchmark³², extends assessment into tandem repeats and homopolymers, although satellite arrays and many segmental duplications remain excluded.

Here we present LongPhase 2, which jointly phases sequence variation and allele-specific methylation, accounts for sequence context and identifies potentially unreliable phase assignments. It places 5mC sites in the same weighted phasing graph as SNVs, small indels and SVs, letting methylation inform the phase while it is being built. It weights every read observation by base quality and sequence context, giving less weight to homopolymers, tandem repeats and copy-number-altered regions. It scores the phasing uncertainty of every variant, and a graph neural network, enabled by default, identifies phase-disagreement onsets for selective unphasing, leaving the genotypes unchanged. We evaluate it against both GIAB v4.2.1 and the T2T-HG002-derived benchmark, and it keeps the command-line interface and output formats of LongPhase.

Results

LongPhase 2 combines joint phasing with learned error detection

LongPhase 2 builds one phasing graph per chromosome from SNVs, small indels, SVs and allele-specific 5mC sites called by its `modcall` module (Fig. 1a). Read observations are weighted by base quality and sequence context, and filters reduce the influence of observations in homopolymers, tandem repeats and candidate copy-number-altered regions (Fig. 1b). Every phased variant then receives a phasing-uncertainty score, and a graph neural network (GNN) examines windows centred on high-uncertainty sites. Predicted phase-disagreement onsets are selectively unphased, with the genotypes retained, and blocks disconnected by the removal are split (Fig. 1c); we refer to this step as GNN correction. The output comprises phased VCFs for each variant class and haplotagged alignments (Fig. 1d).

We evaluated LongPhase 2 on HG002 nanopore R10.4.1 reads down-sampled to 10–60× and on PacBio HiFi reads from the same individual at 10–50× (Methods). Nanopore data were sampled in 2× steps from 10 to 20×, with ten replicates per coverage, and once each at 30,



Figure 1: **Overview of LongPhase 2.** **a**, SNVs, small indels, SVs and 5mC sites carried by the same long reads enter one phasing problem; alleles are coloured by read haplotype. **b**, Weighted phasing graph. Read-supported edges link allele nodes, and each observation is weighted by base quality and sequence context; the inset shows an indel down-weighted inside a tandem repeat. **c**, Each variant receives a phasing-uncertainty score (Methods), and a graph neural network (GATv2 and Transformer layers; Supplementary Fig. 6) classifies the variants around the most uncertain ones. Predicted phase-disagreement onsets are selectively unphased (grey); we call this step GNN correction. **d**, Output: phased haplotypes of all four variant classes, with the selected variant left unphased (faded), and haplotagging reads.

40, 50 and 60×; HiFi data and the analyses of unphased variants used one down-sampling per coverage. All phasers received the same alignments and variant calls. Phasing was scored against two HG002 truth sets, each used in full, including variants outside its benchmark regions: GIAB v4.2.1, integrated from mapping-based call sets, and GIAB v5.0q, in which every heterozygous variant is phased by the diploid telomere-to-telomere (T2T) assembly (Methods). Unless stated otherwise, results refer to SNV phasing against v5.0q, and LongPhase 2 denotes the complete method, including GNN correction. Per-run values are provided in Supplementary Data 1 and summaries in Supplementary Tables 7–17.

Comparison with existing phasers

LongPhase 2 has the lowest SNV switch error rate at every coverage. In SNV-only phasing, LongPhase 2 had a lower switch error rate than WhatsHap and HapCUT2 at each of the ten coverages, decreasing from 0.082% at 10× to 0.023% at 60× (Fig. 2a). The separation widened between 10 and 20× and then remained stable: the rate of LongPhase 2 was 2.8- and 3.3-fold lower than those of WhatsHap and HapCUT2 at 10×, and 3.4–3.7- and 4.3–4.8-fold lower at 20–60× (Supplementary Table 8). The difference held in every down-sampling replicate: compared within the same replicate at 10–20×, the rate of LongPhase 2 was 2.8–4.1-fold lower than that of WhatsHap and 3.2–5.1-fold lower than that of HapCUT2. The separation differed between genomic regions and was largest in the regions that v5.0q adds to the GIAB benchmark (Fig. 3d,e). LongPhase 2 also had the lowest block-wise Hamming distance at every coverage, 1.3–1.8-fold below those of the other tools (Fig. 2b).

The lower error rates were not obtained with shorter blocks (Fig. 2c). Block N50 increased with coverage at a similar rate for the three tools, from 0.84–0.95 Mb at 10× to 2.72–2.86 Mb at 60×, so that the path of LongPhase 2 lies at lower switch error rates than those of the other tools over the whole coverage range. Its blocks were up to 11% shorter than those of WhatsHap at 10–20× and 1–5% longer from 30×. The shortfall at low coverage arises from GNN correction, which splits blocks where it unphases variants: before correction, LongPhase 2 already had 1.9–2.5-fold lower switch error rates than WhatsHap, with block N50 between 3% shorter and 7% longer at every coverage (Supplementary Table 8). When indels, SVs and 5mC were phased together with SNVs, LongPhase 2 built blocks 6–50% longer than those of WhatsHap at every coverage while keeping 2.5–3.1-fold lower SNV switch error rates (Supplementary Tables 8 and 10). The fractions of benchmark variants phased by each tool are compared below, together with the calls that each tool leaves unphased (Fig. 4).

The advantage did not depend on the chromosomes used to train the GNN. On chr17, chr21 and chr22, which were held out from training, every tool had higher switch error rates than genome-wide, and LongPhase 2 had 2.8–3.9-fold lower rates than WhatsHap and 3.3–4.9-fold lower rates than HapCUT2 at 10–20× (2.7–4.9-fold at 30–60×, from single runs with 37–83 switch errors by LongPhase 2; Supplementary Table 15).

Of two further phasers run at 10–20× (Supplementary Fig. 10), Ralphi²⁷ had 2.5–3.5-fold higher switch error rates than LongPhase 2. Margin⁹ had the lowest rate of all tools at 10× (0.069% versus 0.082%) and the same rate as LongPhase 2 at 12×, but its rate decreased little with coverage and was 1.2–1.5-fold higher than that of LongPhase 2 at 14–20×; its blocks were

2.1–3.2-fold shorter at every coverage. The held-out chromosomes showed the same trade-off: Margin had lower rates than LongPhase 2 at 10 and 12× (0.099% versus 0.127% at 10×), with 3.3-fold shorter blocks (Supplementary Table 15). GCphase was not evaluated because it treats every input record as a heterozygous SNV (Methods).

LongPhase 2 was also faster (Supplementary Table 16). On one thread, SNV phasing with GNN correction took 8.1 min at 10× and 18.0 min at 60×, compared with 25.2 and 113.8 min for WhatsHap and 24.1 and 96.4 min for HapCUT2; between these coverages, its runtime increased 2.2-fold, compared with 4.0–4.5-fold for the other tools. At 60×, its peak memory was 7.4 GiB on one thread and 63.0 GiB with 24 threads (16.3 min), compared with 4.0 GiB for WhatsHap and 19.4 GiB for HapCUT2.

The advantage of LongPhase 2 extends to indels and increases with coverage. WhatsHap can also phase indels with SNVs; doing so lengthened its blocks by 31–59% but raised its SNV switch error rate by 17–20% at every coverage (Fig. 2c,f). With the same indels, LongPhase 2 kept 3.0–3.7-fold lower SNV switch error rates than WhatsHap (Fig. 2d, solid lines), and the difference was larger for the indels themselves (dotted lines). For consecutive variant pairs that included an indel, the switch error rate of LongPhase 2 decreased with coverage from 0.101% to 0.030%, 8–25% above its SNV rate, whereas that of WhatsHap remained at 0.75–0.91%, three- to ninefold above its SNV rate. The difference therefore grew with coverage, from 8-fold at 10× to 31-fold at 60× (Supplementary Table 14). The indel Hamming distance followed the same pattern, decreasing from 6.8% to 3.5% for LongPhase 2 but remaining at 9.0–13.8% for WhatsHap (Fig. 2e). WhatsHap built longer blocks from SNVs and indels, although this difference narrowed from 28–34% at 10–18× to 14% at 60× (Fig. 2f). When LongPhase 2 also co-phased SVs and 5mC, which WhatsHap cannot use, its blocks were 2–21% shorter than those of WhatsHap with indels, and 2–3% shorter at 50–60×, at 3.0–3.7-fold lower SNV switch error rates (0.027% versus 0.097% at 60×; Fig. 2f, dashed line).

Joint SNV and 5mC phasing makes fewer switch errors than MethPhaser. MethPhaser²¹ adds methylation evidence to existing SNV phase blocks; we gave it the SNV-only output of LongPhase 2 without GNN correction (Methods). Joint SNV and 5mC phasing by LongPhase 2 had 1.5–1.7-fold lower switch error rates than MethPhaser at every coverage (Fig. 2g), and its Hamming distance was 1.4-fold lower at 10× and 2.0–2.9-fold lower at 20–60× (Fig. 2h). [Block N50 against MethPhaser (Fig. 2i and Supplementary Fig. 11d): pending issue #5, MethPhaser blocks overlap on 3–4 chromosomes per coverage.] The phased fractions agreed to within 0.2 percentage points (Supplementary Fig. 11b). MethPhaser joined blocks of its input without changing the phase within them; these joins added 2–15% to the switch errors of its input and raised its Hamming distance by 4% at 10× and 1.8–2.8-fold at 20–60×. Before GNN correction, joint phasing made as many switch errors as the SNV-only input of MethPhaser (within 2%; Supplementary Fig. 11). GNN correction therefore accounted for 63–95% of the difference in switch errors in Fig. 2g, and the joins made by MethPhaser for most of the remainder.

The advantage of LongPhase 2 carries over to PacBio HiFi data without retraining. On HiFi data, with the network trained on nanopore data, LongPhase 2 had 2.2–2.6-fold lower

SNV switch error rates than WhatsHap at every coverage (Fig. 2j and Supplementary Table 11). Because the rate of WhatsHap decreased more steeply with coverage (from 0.128% to 0.077% between 10 and 50 $\times$, versus 0.052% to 0.034%), the difference narrowed from 2.5–2.6-fold at 10–30 $\times$ to 2.2–2.3-fold at 40–50 $\times$. Hamming distances were 0.7–1.2% for both tools; that of LongPhase 2 was lower at 10–40 $\times$ and higher at 50 $\times$ (Fig. 2k). Block N50 increased from 340 to 580 kb and differed by at most 3% between the tools (Fig. 2l). HiFi blocks were shorter than nanopore blocks, consistent with the shorter HiFi reads, and the gap widened from 2.5-fold at 10 $\times$ to 4.3-fold at 50 $\times$ because block N50 increased 1.7-fold over this range on HiFi data but 3.0-fold on nanopore data.

Benchmark choice changes the measured separation between phasers. Against GIAB v4.2.1, the three phasers converged as coverage increased (Fig. 3a). LongPhase 2 made 1,635–1,820 switch errors at every coverage, whereas the switch errors of WhatsHap and HapCUT2 decreased from 2,773 and 3,050 at 10 $\times$ to 1,801 and 2,088 at 60 $\times$, so the ratio between the highest and the lowest rate decreased from 1.7 to 1.2–1.3 at 30–60 $\times$. Against v5.0q, the same phased VCFs did not converge (Fig. 3b): the switch errors of every tool decreased with coverage, those of LongPhase 2 from 1,518 to 496, and the ratio increased from 3.3 to 4.3–4.8 (Supplementary Table 9). Relative to v4.2.1, v5.0q lowered the switch errors of LongPhase 2 by 17% at 10 $\times$ and by 70% at 60 $\times$, whereas it raised those of HapCUT2 at every coverage and those of WhatsHap below 60 $\times$ (both 1.6-fold at 10 $\times$). Hamming distance changed little between the truth sets.

Against v4.2.1, the tools made an increasing share of their switch errors at the same SNV pairs as coverage increased. The share of the switch errors of LongPhase 2 that coincided with switch errors of both other tools rose from 63% at 10 $\times$ to 93% at 60 $\times$, where the same set accounted for 86% of the switch errors of WhatsHap and 75% of those of HapCUT2; against v5.0q, the corresponding shares at 60 $\times$ were 37%, 10% and 8% (Fig. 3c and Supplementary Table 13). Within the regions assessed by both truth sets, each tool made 1,127–1,492 switch errors against v4.2.1 at 30–60 $\times$ but 32–393 against v5.0q, so the two truth sets differ in phase at many sites there. Whether this reflects errors in the v4.2.1 phase transfer has not been examined in the reads.

Within the v5.0q benchmark regions, LongPhase 2 had the lowest switch error rate at every coverage, and the separation increased with coverage, from 2.1-fold relative to WhatsHap and 2.6-fold relative to HapCUT2 at 10 $\times$ to 3.5–4.0-fold and 8.0–10-fold at 30–60 $\times$ (Supplementary Table 13). Among GIAB genomic strata²⁵, the separation was largest in segmental duplications (4.3–6.5-fold at 10 and 60 $\times$) and smallest outside difficult regions (1.7–2.0-fold; Fig. 3d,e). The separation was largest in the v5.0q benchmark regions that v4.2.1 does not cover, which v5.0q adds to the benchmark³²: there, WhatsHap and HapCUT2 had 5.6–9.3 and 7.5–12 times the switch error rate of LongPhase 2 at every coverage (15–41 switch errors for LongPhase 2), compared with 1.7–2.6 and 2.1–9.8 times in the regions that both truth sets cover (Fig. 3d,e and Supplementary Table 13). As coverage increased, a growing share of the switch errors of every tool lay outside the benchmark regions of both truth sets, where the truth phase is least certain: 29% at 10 $\times$ and 70% at 60 $\times$ for LongPhase 2, and 43–46% and 62–75% for the other tools. There, LongPhase 2 had switch error rates of 0.8–1.2% and the other tools 2.6–4.9%. These sites accounted for 50–56% of the genome-wide difference between LongPhase 2 and the other

tools at 10 $\times$ and for 60–77% at 60 $\times$; we regard this part of the comparison as exploratory.

LongPhase 2 phases slightly fewer benchmark SNVs, and more of the calls it phases are benchmark SNVs. All three tools received the same heterozygous SNV calls. WhatsHap and HapCUT2 phased 0.7–1.0 percentage points more of the benchmark heterozygous SNVs than LongPhase 2 at every coverage; for all three tools, the phased fraction rose steeply up to 16 $\times$ and levelled off at 91–92% from 20 $\times$ (Fig. 4a and Supplementary Table 8). Most of this difference was present before GNN correction (0.6–0.9 percentage points), and correction added 0.1–0.3. On HiFi data, WhatsHap phased 0.4–0.5 percentage points more at every coverage, and the phased fraction of both tools rose from 87% at 10 $\times$ to 92% at 20–50 $\times$ (Fig. 4b and Supplementary Table 11). In co-phasing with indels, WhatsHap phased 2.4–4.5 percentage points more benchmark heterozygous indels, a difference that increased with coverage (Supplementary Table 9), but its switch error rate at variant pairs that included an indel was 8–31 times that of LongPhase 2 (Fig. 2d). These fractions count only phased calls that match a benchmark heterozygous variant. Of all the heterozygous SNV calls on chr1–22 that each tool phased, the share that matched a benchmark heterozygous SNV rose with coverage, from 89.4% to 93.1% for LongPhase 2 and from 86.2% to 91.5% for WhatsHap (Fig. 4c and Supplementary Table 8). It was higher for LongPhase 2 at every coverage and in every replicate; the difference narrowed from 3.2 percentage points at 10 $\times$ to 1.5 at 50–60 $\times$. The F1 score, the harmonic mean of this share and the phased fraction, was also higher for LongPhase 2 in every replicate, by 0.8–1.0 percentage points at 10–18 $\times$, decreasing to 0.3 at 50–60 $\times$ (83.2–92.2% for LongPhase 2 and 82.3–91.9% for WhatsHap; Supplementary Table 8).

WhatsHap writes more genotype errors onto haplotypes. Genotype errors are one of the error classes that variant benchmarks report³². When a phaser phases a heterozygous call at a position where HG002 is homozygous, it puts a wrong allele on one haplotype. WhatsHap phased 27,135, 13,706 and 11,815 such calls at 10, 30 and 60 $\times$ (positions with a homozygous v5.0q genotype, chr1–22), and LongPhase 2 phased 14,444, 9,828 and 8,497, that is, 1.4–1.9 times fewer (Fig. 4d). The same held for calls with another allele in v5.0q (WhatsHap phased 1.3–1.4 times as many) and for calls without a v5.0q record (1.3–1.4 times; 190,344 versus 152,798 at 60 $\times$). These differences come from the calls that only one tool phased. WhatsHap phased 122,000–149,000 calls that LongPhase 2 left unphased; 15–20% of them matched a benchmark heterozygous SNV, and WhatsHap phased 22.9%, 15.9% and 10.0% of these wrongly, three to six times its rate over all its phased SNVs. LongPhase 2 phased 38,000–54,000 calls that WhatsHap left unphased. About 90% of them were absent from the benchmark, and 92–93% lay in segmental duplications or outside one-to-one assembly alignments, where the benchmark does not assess phase; LongPhase 2 phased 13.9–16.7% of the 1,002–1,579 benchmark heterozygous SNVs among them wrongly. The calls phased by one tool only also lay inside that tool’s own switch-error intervals 2.0–9.6 times more often than its phased calls overall (Supplementary Table 17).

WhatsHap leaves unphased mainly calls with few reads, and LongPhase 2 also calls with excess reads. The calls left unphased by one tool only differed in read depth

(Supplementary Fig. 12). At 10 $\times$, 64–71% of the calls in each set had a depth below 20 $\times$. At 30 and 60 $\times$, this held for 53–60% of the calls left unphased by WhatsHap alone but for only 12–16% of those left unphased by LongPhase 2 alone, whose depths peaked at the sequencing coverage. At 60 $\times$, 29% of the calls left unphased by LongPhase 2 alone had a depth of 90 $\times$ or more, and 15.6% were at or above the depth cap of the variant caller, compared with 3% of those left unphased by WhatsHap alone. At 30 and 60 $\times$, 28% and 17% of the calls left unphased by LongPhase 2 alone had at least twice the sequencing coverage, the depth expected when the reads of two paralogous copies map to one locus, compared with 7% and 2% of those left unphased by WhatsHap alone; such mismapping is a known source of false variant calls in segmental duplications³². Inside the v5.0q benchmark regions, 92.0% of the 3,684 such high-depth calls had no benchmark variant and were therefore false-positive calls. The source of the excess depth was not determined: the calls left unphased by LongPhase 2 alone lay in segmental duplications or outside one-to-one assembly alignments less often than those left unphased by WhatsHap alone (54–75% versus 92–93%).

SNVs unphased by GNN correction lie mostly outside the benchmark and outside one-to-one assembly alignments. GNN correction unphased 55,756–58,162 SNVs per genome at 10–60 $\times$ (one replicate per coverage). They account for 15–18% of the calls that LongPhase 2 alone left unphased at 10, 30 and 60 $\times$; the rest were left unphased before correction. Classified by position against the v5.0q benchmark VCF, the share absent from it increased with coverage from 83.0% at 10 $\times$ to 93.0% at 60 $\times$, and the share heterozygous in it decreased from 11.9% (6,650) to 4.0% (2,215); the remainder were homozygous (Fig. 4e). Requiring matching REF and ALT alleles left 6,405 and 2,055 heterozygous benchmark SNVs. Absence from the benchmark VCF does not by itself establish that a call is erroneous, because the benchmark depends on assembly alignment and variant representation.

In an exploratory analysis of the T2T-HG002 assembly alignments, 11.8% of the 56,076 SNVs unphased at 60 $\times$ lay in regions where both assembled haplotypes aligned one-to-one to GRCh38, compared with 92.6% of a random sample of SNVs phased before correction (odds ratio for lying outside these regions, 94; Fig. 4f and Supplementary Table 12). These counts include chrX and chrY; on the autosomes alone, the shares were 9.6% and 92.9% (odds ratio, 124), and the enrichment was observed on every autosome (Fig. 4g). Sites outside these regions clustered in pericentromeric satellites, particularly at chr11:51–54 Mb, chr12:35–36 Mb and chr18:15–20 Mb. Of the 6,601 unphased SNVs within one-to-one alignments, 3,421 (52%) had no assembly variant record at the same start coordinate; because this test does not compare alleles, it can miss variants that the assembly represents at another position. The analysis has three further limits. The background sample included sites that GNN correction later unphased, which biases the comparison towards no difference. The one-to-one regions come from the raw assembly alignment, which retains assembly errors that the benchmark curation removes. Finally, because v5.0q derives from the same assembly, the analysis extends the assessment beyond the benchmark regions but does not independently validate phase or allele identity.

Contribution of the internal modules

GNN correction removes one-third of SNV switch errors. Before GNN correction, LongPhase 2 already had 1.9–2.5-fold lower SNV switch error rates than WhatsHap and 2.2–3.3-fold lower rates than HapCUT2 (Supplementary Fig. 13b and Supplementary Table 8). GNN correction lowered these rates by a further 31–33% at every coverage (30–36% in individual replicates at 10–20×), at a cost that decreased with coverage: the phased fraction fell by 0.26 percentage points at 10× and by 0.09–0.11 at 30–60×, and block N50 by 5–9% at 10–20× and 2–5% at 30–60× (Supplementary Fig. 13a,d). On the held-out chromosomes, GNN correction removed 20–23% of switch errors at 10–20× and 14–20% at 30–60× (9–15 per single run), and it withheld 10.5 phased SNVs per removed switch error, compared with 8.3 over the whole genome (Supplementary Table 15). Because these chromosomes were harder to phase for every tool, the smaller reduction may reflect both sequence not seen in training and chromosome difficulty. On HiFi data, GNN correction removed 19–25% of switch errors and reduced the phased fraction by at most 0.1 percentage points and block N50 by 0.6–1.6% (Supplementary Fig. 13e,f).

GNN correction reduces the additional error of indel co-phasing. We evaluated seven combinations of variant classes, each with and without GNN correction (Supplementary Figs. 14 and 15 and Supplementary Table 10). Without correction, adding SVs increased block N50 by 1–2% and adding 5mC by 2–13%, and neither changed the SNV switch error rate by more than 3%. Adding indels increased block N50 by 29–51% (32–75% together with SVs and 5mC) but also raised the SNV switch error rate by 24–34% at every coverage. GNN correction reduced switch error rates by 30–33% in configurations without indels and by 34–42% in those with indels, removing 41–71% of the additional error associated with indels and 24–39% of the switch errors at indel pairs (Supplementary Table 14). It shortened blocks by 12–22% in indel-containing configurations, compared with 1–9% in the others, as unphasing disconnected blocks that indels had bridged.

After GNN correction, each added evidence class still lengthened blocks at every coverage, and the classes kept their distinct effects on accuracy (Supplementary Fig. 15e–h and Supplementary Table 10). Adding 5mC kept the SNV switch error rate within 4% of that of SNV-only phasing and increased block N50 by 2% at 10× and by 8–14% from 18×. Adding indels increased block N50 by 15–32% but also raised the switch error rate by 12–21%. Phasing all four classes gave the longest blocks at every coverage, 19–47% longer than those of SNV-only phasing (4.10 versus 2.86 Mb at 60×), at 14–27% higher switch error rates; adding SVs and 5mC to indels changed the switch error rate at indel pairs by at most 6% (Supplementary Table 14). Every configuration had a 2.5-fold or more lower SNV switch error rate than WhatsHap with or without indels at the same coverage (Supplementary Tables 8 and 10). In four-class phasing before GNN correction, LongPhase 2 phased 30.8–45.5% of benchmark heterozygous indels, 99.0–99.4% of the heterozygous 5mC sites called by `modcall` and 68.1–70.1% of the heterozygous SVs called by `Sniffles2`; GNN correction then unphased 1.4–4.6% of the phased 5mC sites and 2.4–3.8% of the phased SVs[Add the phased SV and 5mC counts per configuration and coverage (audit table `q10_sv_mod.tsv`) to Supplementary Data 1].

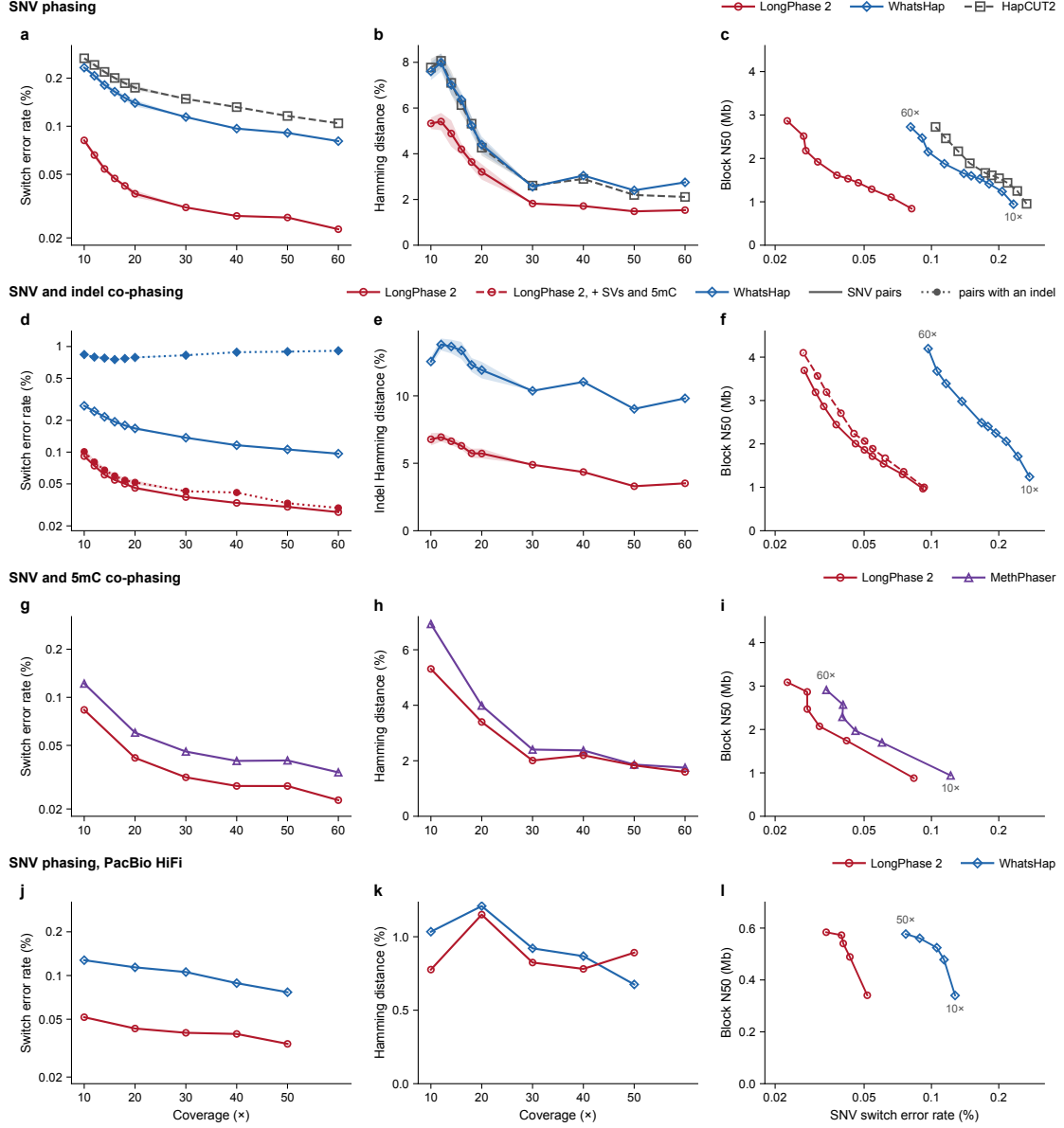


Figure 2: Accuracy and contiguity of LongPhase 2 compared with other phasers for each type of phasing evidence and on PacBio HiFi data. HG002 nanopore R10.4.1 reads down-sampled to 10–60× (**a–i**) and PacBio HiFi reads down-sampled to 10–50× (**j–l**); identical variant calls for every tool, scored against the T2T-HG002-derived v5.0q benchmark. Nanopore values at 10–20× are means of ten down-sampling replicates with the s.d. as a light band; all other values come from one down-sampling (Methods). Columns show the switch error rate (log scale), block-wise Hamming distance, and SNV switch error rate against phase-block N50, one point per coverage from the lowest (bottom right) to the highest (top left). Phased fractions are shown in Fig. 4. **a–c**, SNV phasing compared with WhatsHap and HapCUT2. **d–f**, SNV and indel co-phasing compared with WhatsHap; in **f**, the dashed line adds SVs and 5mC, which WhatsHap cannot use, to the co-phasing of LongPhase 2. In **d**, solid lines score consecutive SNV pairs and dotted lines consecutive pairs that include an indel; **e** refers to indels (Supplementary Table 14). **g–i**, SNV and 5mC co-phasing compared with MethPhaser, replicate 1; MethPhaser was given the SNV-only output of LongPhase 2 without GNN correction (Methods). **j–l**, SNV phasing of PacBio HiFi (Revio) data compared with WhatsHap, with variants called at each coverage; LongPhase 2 uses the network trained on nanopore data. The N50 axis of **l** differs from those of **c**, **f** and **i**.

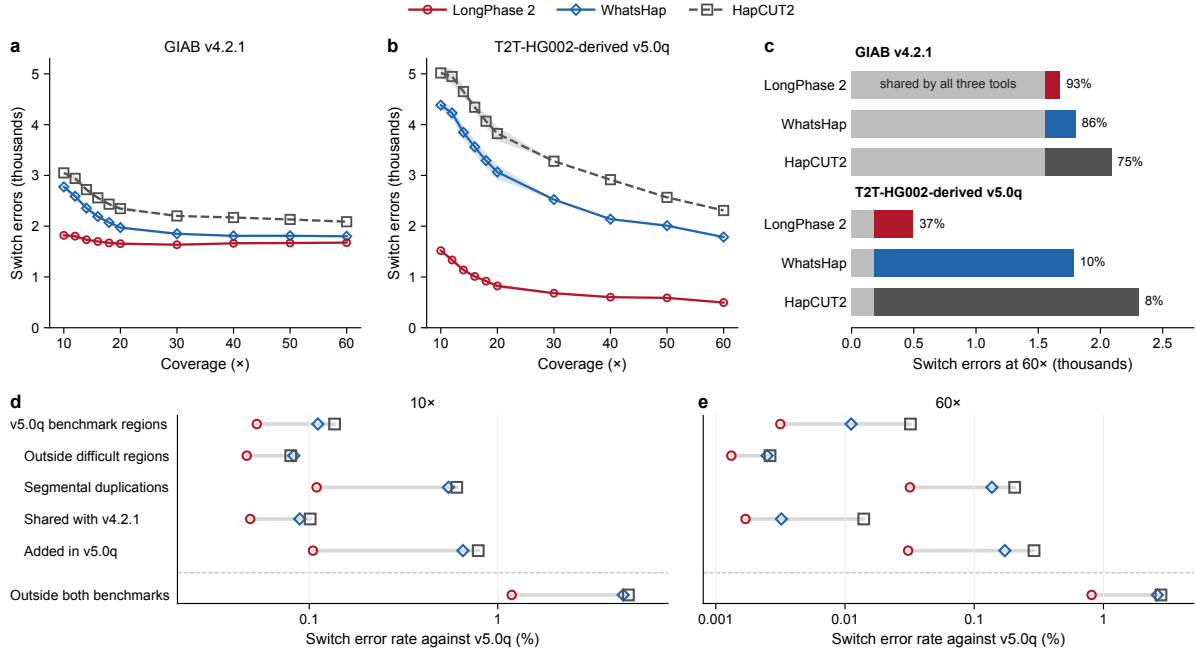


Figure 3: The same phased VCFs scored against GIAB v4.2.1 and the T2T-HG002-derived v5.0q benchmark. SNV-only phasing of HG002 nanopore R10.4.1 data; replicates as in Fig. 2; both truth VCFs used in full. **a,b**, Genome-wide switch errors under v4.2.1 (**a**) and v5.0q (**b**). **c**, Switch errors at 60x; grey, switch errors made by all three tools at the same SNV pair, with their share of each tool's total. **d,e**, Switch error rate against v5.0q at 10x (**d**) and 60x (**e**) inside the v5.0q benchmark regions and in subsets of them: outside difficult regions and in segmental duplications (GIAB v3.6 strata), and in the parts that v4.2.1 also covers or that v5.0q adds. Below the dashed line, outside the benchmark regions of both truth sets, where the truth phase is least certain (exploratory). A variant pair is assigned to a region when both of its variants lie in it (Supplementary Table 13).

Calls phased by LongPhase 2 and WhatsHap

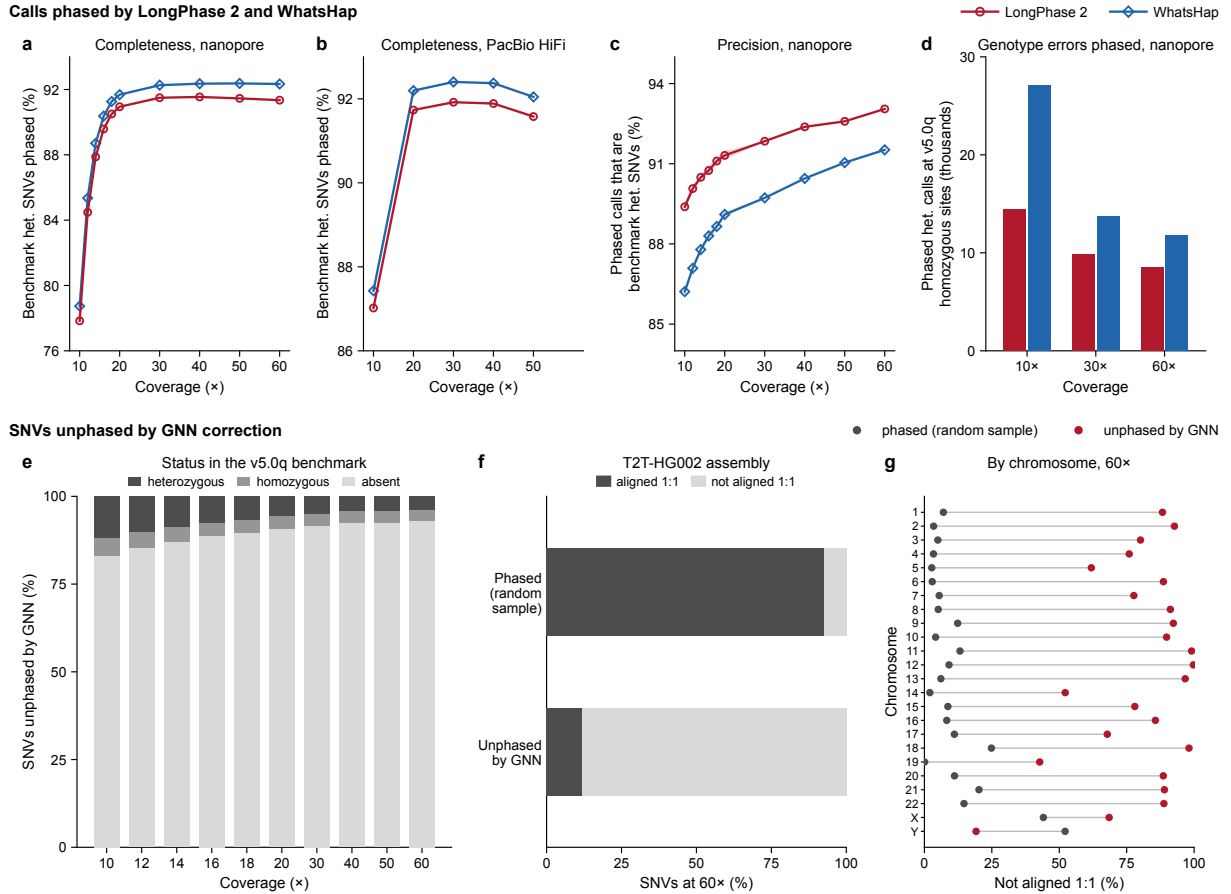


Figure 4: Heterozygous SNV calls phased by LongPhase 2 and WhatsHap, and SNVs unphased by GNN correction. SNV-only phasing of HG002 nanopore R10.4.1 data (**a**, **c–g**) and PacBio HiFi data (**b**). **a–c**, coverages and replicates as in Fig. 2; **d–g**, replicate 1. **a,b**, Fraction of the heterozygous SNVs of the v5.0q benchmark phased from nanopore (**a**) and HiFi (**b**) data; both tools received the same heterozygous calls. **c**, Share of the phased calls on chr1–22 that match a benchmark heterozygous SNV in position and alleles. **d**, Phased heterozygous calls on chr1–22 at positions where v5.0q has a homozygous genotype (genotype errors; Supplementary Table 17). **e**, SNVs phased before and unphased after GNN correction, by their genotype in the v5.0q benchmark VCF, matched by position without a benchmark BED. **f**, At 60x, these SNVs and a random sample of SNVs phased before correction, inside or outside the regions where both haplotypes of the T2T-HG002 assembly align one-to-one to GRCh38 (exploratory). **g**, The share outside these regions for each chromosome (Supplementary Table 12).

Discussion

LongPhase 2 integrates genetic variation and allele-specific methylation in a shared phasing graph, then withholds phase assignments predicted to be unreliable. Joint use of these evidence classes extends earlier approaches that use methylation or SVs to connect haplotype blocks in regions with few heterozygous SNVs^{2,11,20,21,33}. Whereas LongHap uses methylation to bridge blocks after sequence phasing²³, LongPhase 2 incorporates methylation and sequence evidence during haplotype construction. The evidence classes made distinct contributions: methylation increased contiguity at similar SNV switch error rates, whereas indels accounted for most of the increase in both contiguity and error. The corresponding trade-off in WhatsHap suggests that this limitation is not specific to LongPhase 2, although shared difficulties in representing or modelling indels may contribute. The indels themselves were phased with 8–31-fold fewer switch errors than by WhatsHap. GNN correction reduced the additional error associated with indels, and improvements in indel calling could further improve this balance^{15,34}.

Selective withholding complements evidence integration by reducing reliance on uncertain phase assignments. LongPhase 2 reports phasing entropy and weighted vote totals as measures of the underlying evidence. These differ from the model-based confidence estimates provided by HapCUT2⁸ and statistical phasers⁵: when votes are present, low entropy indicates their agreement, not necessarily strong evidence or a low probability of error. The GNN incorporates neighbouring graph and sequence features, which may allow it to identify candidates that a high-entropy threshold alone would miss. However, its added value over simpler withholding rules and HapCUT2’s confidence pruning remains to be established at matched variant retention. Its onset-prediction score is neither calibrated nor included in the output. Because withholding variants and splitting blocks change which variant pairs can be assessed, reductions in switch error rate must be interpreted jointly with phased fraction and contiguity. Before GNN correction, LongPhase 2 already made 1.9- to 2.5-fold fewer switch errors than WhatsHap at similar block N50, and most of the additional SNV calls that WhatsHap phased did not match a benchmark heterozygous SNV. Of the benchmark heterozygous SNVs that only WhatsHap phased, it phased 10–23% wrongly, three to six times its overall rate; LongPhase 2 erred at a similar rate at the 13–25 times fewer SNVs that only it phased. Margin’s lower error rate but shorter blocks at 10× genome-wide, and at 10–12× on the held-out chromosomes, illustrate this trade-off. Most SNVs unphased by the GNN lay outside regions with one-to-one assembly alignments, where the present analysis cannot establish whether their phase was correct.

Benchmark choice substantially altered the measured separation between methods. At 30–60×, the highest and lowest switch error rates differed by 1.2–1.3-fold against GIAB v4.2.1 but by 4.3–4.8-fold against v5.0q. Two effects contributed. Against v4.2.1 at high coverage, most switch errors of the three tools fell at the same variant pairs. In regions that both benchmarks assess, the same phased outputs had 1,127–1,492 switch errors against v4.2.1 but 32–393 against v5.0q at 30–60×, which indicates that the two truth sets differ in phase there. All three phasers agreed with v5.0q at most of these sites, including WhatsHap and HapCUT2, which use no parameters trained on either truth set. Errors in the v4.2.1 phase transfer are one possible explanation. The phasers share reads, alignments and variant calls, however, and we did not check these sites independently, for example in the reads or with parental data. Within the

v5.0q benchmark regions, LongPhase 2 made 2.1- to 10-fold fewer switch errors than the other phasers. Most of the remaining genome-wide separation lay at assembly-phased variants outside the benchmark regions of both truth sets, where the truth phase is least certain; this part of the comparison is exploratory, and scoring outside the benchmark regions is a limitation of our evaluation. The v5.0q release evaluated variant calls but not phase³². In our data, the phasers differed most in the regions that v5.0q adds to the benchmark, and the choice of which calls to phase changed how many genotype errors were written onto haplotypes, an effect that variant benchmarks do not measure. Our results indicate that phasers should be compared against assembly-phased truth sets, with errors reported inside and outside the benchmark regions and stratified by genomic context²⁵.

The current evaluation has several limits. All data derive from HG002, and the network was trained on nanopore runs from the same down-sampling series, using labels from the evaluation benchmark. Genome-wide metrics include the 17 chromosomes that contributed training windows. On the three held-out chromosomes, the advantage over WhatsHap and HapCUT2 was unchanged, but GNN correction removed fewer switch errors (14–23% versus 31–33% genome-wide) at a higher cost per removed error (10.5 versus 8.3 withheld SNVs), and these results measure performance on chromosomes excluded from network training. These chromosomes were held out from network training but not from the earlier tuning of the graph constants, and they come from the same individual. Evaluation in other individuals against an independent benchmark, with weights and thresholds fixed in advance, is still needed. The HiFi results without retraining are consistent with transfer between sequencing platforms within HG002, but do not establish generalization to other individuals. Subsampling variability was not estimated for nanopore coverages above 20× or for HiFi. For SVs and 5mC, we measured phased fractions rather than phasing accuracy, and these sites had no supervised training labels. Methylation co-phasing was assessed only on autosomes of a male lymphoblastoid cell line²¹. The method also assumes a diploid genome with balanced allele fractions.

Compatibility with pipelines that use LongPhase^{14,16,18,19} enables evaluation of the downstream effects of selective phase withholding. In haplotype-aware somatic calling^{16,19}, reducing errors in germline phase could improve the assignment of tumour reads to haplotypes, but withholding phase may also reduce the information available to the caller. Varying the germline phaser while keeping the somatic caller fixed would quantify this trade-off. A calibrated per-variant confidence measure would further allow downstream methods to account explicitly for uncertainty in inherited phase.

Methods

Overview of LongPhase 2

LongPhase 2 is a diploid germline phaser that takes long-read alignments, a reference genome and heterozygous variant calls, and returns phased VCFs and haplotagged alignments (Fig. 1). SNVs, small indels, structural variants (SVs) and allele-specific 5-methylcytosine (5mC) observations are represented together in one weighted graph per chromosome. Initial haplotypes are resolved from weighted multi-neighbour votes, reads are used for a second phase-consistency pass, and a

graph neural network (GNN) then scores candidate phase-disagreement onsets. Selected variants are unphased, never flipped; if this disconnects a haplotype block, the surviving components receive updated phase-set identifiers. The detailed module rules are given in Supplementary Methods 1–6. The method extends the graph formulation of LongPhase 1¹¹.

Input variant calls

Heterozygous SNVs and small indels were taken from PEPPER-Margin-DeepVariant⁹; homozygous and filtered records were carried through unchanged but did not enter the graph. Indels in short tandem repeats were retained as evidence but down-weighted, because a misplaced repeat indel can link the wrong alleles. SVs were taken from Sniffles2¹² or cuteSV¹³ with supporting read names; homozygous SVs, duplicated positions and SVs coincident with SNVs were excluded. The exact eligibility and repeat annotation rules are in Supplementary Method 2.

Calling allele-specific methylation

Modification probabilities were read from the MM/ML tags produced by Dorado³⁵ and carried through alignment. A CpG was called methylated when its 5mC probability was at least 0.8, unmethylated when it was at most 0.2, and ambiguous otherwise. Forward- and reverse-strand observations were mapped to one CpG coordinate. A site was retained as a candidate allele-specific site when methylated and unmethylated reads were sufficiently balanced and the ambiguous fraction was small:

$$\frac{\min(M_R, U_R)}{\max(M_R, U_R)} \geq 0.6, \quad \frac{A_R}{M_R + U_R + A_R} \leq 0.2, \quad (1)$$

where M_R , U_R and A_R are the three read counts. Candidates were then required to co-segregate with a neighbouring heterozygous marker. For a pair of sites, writing RR , RA , AR and AA for the four observed state combinations, linkage required

$$\frac{\max(RR + AA, RA + AR)}{RR + AA + RA + AR} \geq 0.9. \quad (2)$$

The same criterion was used for SNV-anchored and methylation-only linking. Only CpG dinucleotides were considered. Weak candidates were admitted in a later expansion pass, and consecutive CpGs were merged into one locus so that both strands shared one read list. The resulting methylated and unmethylated read names and depths were supplied to the phasing graph. The depth bound and expansion rules are in Supplementary Method 1.

Read-level allele observations and filtering

Alignments with mapping quality below 1, unmapped, secondary or duplicate records were excluded. SNV alleles came from the aligned base and its Phred quality; indel alleles from the CIGAR; SV alleles from a matching CIGAR event or the caller’s read list; and 5mC alleles from the methylation read lists. Supplementary alignments were retained as evidence, so when two alignments of the same read overlapped by more than 20% of their combined span, the shorter alignment was discarded. Under the nanopore setting, a downstream SNV was removed when

two SNVs at most 2 bp apart both lay in homopolymers of length at least three. A copy-number-aware filter identified intervals bounded by excess front- or back-clipping and removed variants whose alternate-supporting reads had an unusually high mismatch load relative to reference-supporting reads. Specifically, a read's mismatch load R_m counted the heterozygous sites in the interval at which it carried the alternate allele, and each variant was scored by

$$\text{MR}(v) = \frac{\bar{R}_m^{\text{ALT}}}{\bar{R}_m^{\text{REF}} + \bar{R}_m^{\text{ALT}}}, \quad (3)$$

where the barred terms are the mean loads of reads supporting the ALT and REF alleles of v . Sites with $\text{MR}(v) \geq 0.7$ were removed from all reads. The three-state interval scanner and its constants are specified in Supplementary Method 2.

Weighted phasing graph

Each heterozygous variant contributed reference and alternate allele nodes. For each read, every allele was connected to the next 35 observed variants, with no edge-distance cap. Variants followed by a gap of more than 300 kb to the next heterozygous variant were skipped during haplotype resolution, casting no votes and receiving no haplotype or entropy. A read contributed weight 1 when both observations had quality at least 12 and weight 0.1 otherwise. Indel and 5mC observations and SV alternate observations were assigned fixed quality 60; reference-allele SV observations were assigned 30. Thus all such observations have full weight at the default threshold, while reference-SV evidence is down-weighted only at a higher threshold.

For a variant pair (u, v) , the four allele-pair supports were accumulated as w_{rr} , w_{ra} , w_{ar} and w_{aa} . The two possible pairings have support $P = w_{rr} + w_{aa}$ and $Q = w_{ra} + w_{ar}$, and their ambiguity is $s = \min(P, Q) / \max(P, Q)$.

Haplotype resolution by multi-neighbour voting

Variants were processed in coordinate order. Each variant cast a vote to each downstream neighbour for the pairing with larger support, except when $s > 0.7$; SNV-5mC pairs used a 0.3 threshold and required at least one read-equivalent of support. Votes had weight 1 by default, 20 for near-unanimous support ($s \leq 0.1$ with total support at least 1, or one pairing below 1 while the other was at least 1), and 0.1 when the voting variant was a tandem-repeat indel. The vote was translated using the already assigned haplotype of the upstream variant, producing totals $h_1(v)$ and $h_2(v)$. When more than three votes had support of at most 1 read-equivalent, and at least one qualifying vote remained, only votes with $s < 0.2$, weight at least 1 and a non-indel voting variant were retained. The larger total assigned the haplotype. A tie beyond the last accepted edge opened a new block; a tie within an existing block left the variant unphased and cast no votes. The first variant position defined the phase-set identifier. The constants were fixed by a pre-study sweep on HG002 at 10-60 $\times$ and were not re-tuned for the reported experiments; the parameter tuning is described in Supplementary Method 3.

Read-based correction and phasing entropy

After the graph assignment, each read was assigned to a haplotype by a weighted majority of its phased SNV and SV alleles (weight 1), indel alleles (weight 0.1) and 5mC alleles (not used for this pass). Reads were retained only with a majority above 0.65 and total evidence above 1. For each variant, the consistency of the two read populations was

$$\rho = \frac{\max(n_1, n_2)}{n_1 + n_2},$$

where n_1 counts haplotype-1 reads carrying REF plus haplotype-2 reads carrying ALT, and n_2 counts the converse. The majority pairing assigned the phase only when $\rho > 0.75$; otherwise the variant was left unphased.

For each phased variant, phasing entropy was calculated from the weighted haplotype totals after the single-read guard:

$$\text{PE}(v) = - \sum_{i \in \{1,2\}} p_i \log_2 p_i, \quad p_i = \frac{h_i}{h_1 + h_2}. \quad (4)$$

It ranges from 0 for unanimous support to 1 bit for equal support. When $h_1 + h_2 = 0$, PE was set to 0 and the case was identified from the accompanying h_1, h_2 fields; this occurs only at the first variant of a block. PE, h_1 and h_2 were written to the VCF, and accepted voting edges with their accumulated weights formed the GNN input graph.

Graph neural network for phase-error detection

Phasing entropy was used to select candidate windows for refinement. Within each window, the GNN combined phasing evidence, genomic context and graph structure to identify phase-disagreement onsets for selective unphasing, as defined by the training labels. Positive targets include isolated disagreements and the first positions of consecutive disagreement runs. The resulting onset scores are not calibrated probabilities.

The GNN operated on merged SNV/indel, SV and 5mC records. A window of at most 41 variants (82 allele nodes) was opened around every phased variant with $\text{PE} \geq 0.8$, extending 20 variants on either side.

Each allele node carried 31 features in four groups: phasing evidence (PE, the genotype and allele indices, the trigger and phase-set indicators, the signed offset from the trigger normalized by the largest absolute offset in the window, the two weighted haplotype totals h_1 and h_2 , and six transforms of them); genomic context, derived from the reference sequence and input variant positions (homopolymer length, GC content, tandem-repeat unit count, local sequence entropy, variant density and distance to the nearest indel); graph structure (out-degree, phase-set size, log offset from the trigger, dispersion of the outgoing edge weights and a bridge-vertex indicator); and four one-hot variant-class indicators. Six edge features summarized vote weight, genomic distance, allele matching and phase-set relations. Edges were restricted to the window, symmetrized and supplemented with mean-feature self-loops. Exact definitions and normalizations are in Supplementary Tables 1–2.

The network used four GPS graph-transformer layers³⁶, each combining a local four-head

GATv2 branch^{37,38} and a global four-head self-attention branch³⁹, followed by residual normalization and a 128–256–128 feed-forward block. The local GATv2 branch aggregates information along edges of the phasing graph, using edge features to weight the contributions of neighbouring allele nodes. The global self-attention branch allows each node to access information from all nodes within the same window, including those without a direct graph connection. Together, these branches allow the model to combine the allele-linkage evidence represented by the graph with patterns of phasing support and genomic context across the window when predicting phase-disagreement onsets. A two-layer classifier produced the two-class softmax output. Exhaustive architecture and inference details are in Supplementary Method 4 and Table 3.

Only variants no further from the trigger than half the largest trigger distance in the window were scored; the same rule defined the nodes used for training loss. Nodes outside that zone supplied context only. Because the zone is defined on genomic distance and not on variant index, the number of variants it holds depends on the local spacing. The two allele-node scores were averaged, followed by averaging across the windows in which the variant lay inside the scoring zone.

A variant with mean score at least 0.30 was unphased while preserving its genotype. Bridge protection was disabled. If removing these variants disconnected a block, the surviving connected components became separate phase sets (Supplementary Method 4).

Training windows came from the same HG002 nanopore co-phasing runs used for evaluation at 10–20× (six coverages and ten down-sampling replicates, 60 runs). Of 722,116 windows, chr1–14 and chr18–20 supplied training (602,788), chr15–16 validation (67,880), and chr17, chr21 and chr22 testing (51,448); the chromosome sets were disjoint. No test-chromosome window entered gradient updates, batch-normalization statistics or checkpoint selection. Windows contained all four variant classes, but only benchmarked SNV and indel nodes had labels, and SV and 5mC correction relies on the model trained on those labelled small variants. The training and evaluation truth was the v5.0q small-variant benchmark. Training windows were limited to 10–20×, whereas results also included 30–60× and PacBio HiFi data.

Labels described local phase orientation. Within each window and phase set, the first benchmarked variant anchored the orientation; later disagreements were labelled as a single-variant flip or unphase event, or as a break at the first site of a longer run. Benchmark-absent variants and SV/5mC records were masked from the loss but retained as graph context. The released model was trained and applied in binary mode, in which flip, unphase and break are merged into a single error class, matching the single action the module is allowed to take. Error labels accounted for 2.96% of labelled central nodes in the training split. Correct nodes were sampled independently in each minibatch at a ratio of at most four per error node, and the resulting cross-entropy used focal weighting⁴⁰ with $\gamma = 1.5$ and label smoothing 0.05. Training used haplotype-swap and coverage-scaling augmentations; the optimizer, stopping rule and environment are listed in Supplementary Table 4.

We used a normalized net-gain criterion for checkpoint selection, assigning unit benefit to each true-positive detection of a label-defined correction target and unit cost to each false-

positive prediction. Checkpoints were selected on chr15–16 by maximizing

$$S = R \left(2 - \frac{1}{P} \right) \quad (5)$$

over 81 score thresholds from 0.01 to 0.99, subject to $R \geq 0.005$, where P and R are precision and recall for the label-defined error class. This expression applies for $P > 0.5$; otherwise the score was $P - 1$. The criterion serves as a proxy for correction utility (Supplementary Method 4). Checkpoint details and classification metrics for validation and held-out chromosomes are reported in Supplementary Table 4.

The deployment threshold $B = 0.30$ was chosen on the 60 HG002 nanopore runs at 10–20× as the threshold that removed the most switch errors while withholding at most ten phased SNVs per removed error. It removed 33.1% of switch errors at 8.3 withheld SNVs per removed error over the whole genome, and 21.2% at 10.5, slightly above the limit, on the held-out chromosomes (Supplementary Method 4 and Supplementary Table 15). **[Provide the per-run threshold-sweep results and the phased-fraction/switch-error trade-off panel.]**

Inference was implemented in C++ without external runtime dependencies; the export procedure and numerical agreement checks are described in Supplementary Method 4.

Haplotagging

Reads were tagged from the phased SNV, indel, SV and 5mC alleles they carried. Each informative observation contributed one vote, and a read was assigned to the majority haplotype when its fraction was at least 0.60; reads whose informative variants belonged to multiple phase sets or lacked a majority were left untagged. Tagged reads received HP, PS and phasing-quality fields. SV alleles were re-derived from the CIGAR and 5mC evidence was matched by read name. Full evidence and quality definitions are in Supplementary Method 5.

Benchmark data

The benchmark used HG002 (NA24385) Oxford Nanopore R10.4.1 reads from the ONT `giab_lsk114_2022.12` release: two PromethION flow cells (PAG65784 and PAG68757), SQK-LSK114, 400 bases per second and 4-kHz sampling, basecalled with Dorado `dna_r10.4.1_e8.2_400bps_sup@v4.0.0` and 5mCG_5hmCG@v2; their read N50 values were 29.2 and 29.4 kb. Reads were aligned to the GRCh38 no-alt analysis set by Oxford Nanopore with `minimap2 v2.24 (-y -ax map-ont)`, retaining reads with mean base quality ≥ 10 . The 60× data set is a random 91.14% subset of the 70.4× alignment (`samtools v1.15, view -s 0.9114`), and the 10–50× data sets were down-sampled from it with `samtools v1.23.141 view -s seed.fraction`. Ten replicates were used at 10–20× and one at 30–60×; variants were called separately at every coverage and replicate. SNVs and indels came from PEPPER-Margin-DeepVariant r0.8, which reports unphased genotypes, SVs from Sniffles2 v2.8.0, and allele-specific 5mC sites from Long-Phase 2. The seeds were 1–9 and 867 at 10× (fraction 0.1666), 12–21 at 12× (0.200), 22–31 at 14× (0.233), 32–41 at 16× (0.266), 42–51 at 18× (0.300), 1–10 at 20× (0.3333), and 42 at 30, 40 and 50× (0.5, 0.6667 and 0.8333). PEPPER-Margin-DeepVariant used the `-ont_r10_q20` model for nanopore data and `-hifi` for HiFi data. HG002 PacBio HiFi (Revio) reads were

taken from a 48× GIAB alignment (Supplementary Table 5), which was used as 50×; 10–40× data sets were down-sampled from it with samtools v1.23.1 `view -subsample` (seed 1; fractions 0.2083, 0.4167, 0.6250 and 0.8333), and variants were called separately at every coverage.

Accuracy was scored with `longphase compare` on autosomes 1–22 against two HG002 truth sets: GIAB v4.2.1²⁹, in the version with phase transferred from a hifiasm assembly (the v4.2.1 benchmark VCF itself is unphased), restricted to biallelic heterozygous records; and GIAB v5.0q³², derived from the diploid T2T-HG002 assembly³⁰, in which every heterozygous variant is phased. The truth VCFs were not restricted to their benchmark regions, and a call was scored only where the truth VCF holds the same REF and ALT alleles in a heterozygous genotype. Both truth VCFs list heterozygous variants outside their benchmark regions (9.3% of the autosomal heterozygous records of v5.0q and 3.4% of v4.2.1), so the error counts include sites that the benchmark regions do not cover, among them satellite and segmental-duplication sequence that the T2T-HG002 assembly resolves³⁰. To locate switch errors and to score indels, every consecutive variant pair in an intersected block was recorded with a re-implementation of `longphase compare`, which reproduced its counts exactly in all 1,024 scored runs. A pair was assigned to a region when both variants lay in it; regions were the intersection and the differences of the two sets of benchmark regions, and the GIAB v3.6 genome stratifications²⁵. Indel switch errors were counted over consecutive pairs that included at least one indel, and the indel Hamming distance was computed in the block orientation chosen on all variants.

Comparison with other methods

WhatsHap v2.8⁷ and HapCUT2 v1.3.4⁸ were run on the same alignments and variant calls; cross-tool SNV comparisons used SNV-only phasing, with indel phasing disabled in WhatsHap. LongPhase 1.0¹¹ and LongPhase 2 were run with the nanopore setting, and GNN correction used PE trigger 0.8, score threshold 0.30, no bridge protection and block splitting. Co-phasing experiments separately evaluated the seven variant-class combinations listed in Supplementary Table 10, with and without correction. Indel co-phasing was also compared with WhatsHap’s default indel phasing. Margin v2.3.1⁹, Ralph²⁷ and MethPhaser²¹ were evaluated as described in Supplementary Table 5. MethPhaser was given the SNV-only output of LongPhase 2 without GNN correction instead of the WhatsHap or HapCUT2 phasing its authors recommend as input, so that MethPhaser and the joint SNV and 5mC phasing of LongPhase 2 started from the same SNV phasing. Without GNN correction, differences between them reflect how methylation is used, after phasing to bridge existing blocks (MethPhaser) or during phasing as additional graph evidence (LongPhase 2); with GNN correction, they also include its effect. GCphase⁴² was not included: it treats every input record as a heterozygous SNV and ignores the FILTER field, so with these call sets it phased about 1.2 million rejected reference calls per 10× genome as heterozygous variants, and a fair comparison would require a different input. The HiFi call sets were phased SNV-only with LongPhase 2 (`-pb`), with and without correction, and with WhatsHap; the GNN weights were trained on nanopore data. WhatsHap was run as `whatshap phase -ignore-read-groups -reference` with `-only-snvs` for SNV comparisons, and HapCUT2 as `extractHAIRS -ont 1` followed by `HAPCUT2` with default parameters, including its default likelihood-based pruning, on the PASS heterozygous SNVs. LongPhase 1.0 was re-run

on every down-sampling replicate. All tools ran on the same workstation.

Evaluation metrics

Metrics (Supplementary Fig. 8) were computed by intersecting heterozygous query and truth records by position and REF/ALT alleles, then partitioning the intersection into blocks phased in both files; singleton blocks were ignored. Within each block, switch errors were the Hamming distance between truth and query switch-encoded haplotype strings, and the switch-error rate divided their number by the assessed consecutive SNV pairs. Isolated switches and two-consecutive-switch flips were reported separately. Hamming distance was the block-wise minimum over the two orientations, divided by the number of compared variants; unlike switch rate, it included indels when present in both files, so it is not comparable between configurations with and without indels, which we therefore compared by SNV switch error rate. Phased fraction was the number of truth heterozygous SNVs or indels that were phased by the query divided by all truth heterozygous variants of that class. Block N50 was the span-weighted median of query phase-set spans of at least two variants, pooled over chromosomes. Switch-error positions were exported for the unphased-SNV analysis. The exact block and denominator definitions are in Supplementary Method 6. [State the exact region-filter commands and add a metric table defining truth eligibility, numerator, denominator and block handling.]

Analysis of unphased SNVs

An SNV was counted as unphased when it was phased in the pre-correction output and unphased after GNN correction. Such sites were classified against the v5.0q benchmark VCF, matched by chromosome and position and without a benchmark BED, as absent, homozygous or heterozygous (Fig. 4e); the heterozygous count was also reported with matching REF and ALT alleles, the rule used by `compare`. The analysis used the 10–60× outputs of replicate 1 with the final GNN model. At 60×, they were also classified against the T2T-HG002 assembly projected to GRCh38 as outside the 1:1 assembly BED, inside without an assembly variant at the same start coordinate, or inside with one; alleles were not compared. All chromosomes, including chrX and chrY, were analysed, and autosome-only values were derived from the per-chromosome counts. A random sample of 50,000 SNVs phased before correction, which included sites that GNN correction later unphased, provided the background; enrichment was summarized by chromosome and 1-Mb bins. To compare tools, the heterozygous SNV calls with single-base alleles and genotype 0/1, which were identical in the two output VCFs, were grouped by the tool that phased them: both tools, LongPhase 2 only, WhatsHap only or neither; calls phased by WhatsHap only were further split by whether LongPhase 2 had already left them unphased before GNN correction. Each call was classified against the v5.0q chr1–22 VCF, without a benchmark BED, as heterozygous with the same position and alleles, heterozygous with other alleles, homozygous or absent; calls on chrX and chrY were counted separately. The calls phased by a tool were those phased by both tools plus those phased by that tool only; phased calls at positions with a homozygous v5.0q record were counted as genotype errors (Fig. 4c). For every SNV-only run of both tools, the share of the phased heterozygous SNV calls on chr1–22 that matched a benchmark heterozygous SNV (precision) was computed from the `compare` count of matched calls,

and the F1 score as the harmonic mean of precision and the phased fraction. For the matched calls phased by one tool only, the phase assigned by that tool was scored as in `compare`, and a call was counted as wrongly phased when its phase was opposite to the majority orientation of its block relative to the truth. A switch-error interval extends from the first SNV of a switch-error pair to the base before the second; the share of the calls phased by one tool only inside that tool's intervals was compared with the share of all heterozygous SNV calls phased by that tool. Calls were also annotated by read depth, by the v5.0q benchmark regions, by GIAB v3.6 segmental duplications and by the one-to-one dipcall alignment (Supplementary Data 1). [Add the script behind Fig. 4e (`analyze_unphased.py`) to the repository; the scripts for Fig. 4c,d are in `notes/unphased`.]

Runtime and memory

Wall-clock time, CPU time and peak resident memory were measured with `/usr/bin/time`; reported LongPhase 2 times included alignment reading, graph construction, phasing, GNN correction and VCF writing, and excluded alignment, variant calling and down-sampling. All runs used one workstation (AMD Ryzen 9 9950X3D, 16 cores and 32 threads, 249 GB of memory, Ubuntu 24.04, input and output on a hard-disk drive). LongPhase 2 was compiled with g++ 13.3 at `-O3` and timed with one and with 24 threads; WhatsHap and HapCUT2 ran single-threaded. Runtime was measured once, on replicate 1 at 10× and on the 60× data set (Supplementary Table 16).

Statistics and reproducibility

Ten nanopore down-sampling replicates at 10–20× were summarized by their mean and standard deviation, while 30–60× and HiFi used one down-sampling per coverage. Subsamples for each platform were drawn from the same source sequencing data set of HG002. The reported standard deviations therefore describe variation due to read subsampling, conditional on that data set, rather than variation among independent sequencing experiments. Subsampling variability was not estimated at 30–60× or for HiFi. Tools compared at each coverage and replicate used the same sampled reads, so fold differences between tools and the effect of GNN correction at 10–20× were also computed within each replicate (Supplementary Table 8). No statistical tests, sample-size calculation, randomization or blinding were used. GNN inference used fixed model weights and no sampling.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data analysed in this study are publicly available: the GRCh38 no-alt analysis set from NCBI (https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/001/405/GCA_000001405).

15_GRCh38/seqs_for_alignment_pipelines.ucsc_ids/); the GIAB HG002 v4.2.1 phase-transfer VCF (https://ftp-trace.ncbi.nlm.nih.gov/giab/ftp/release/AshkenazimTrio/HG002_NA24385_son/NISTv4.2.1/GRCh38/SupplementaryFiles/) and the v5.0q benchmark (https://ftp-trace.ncbi.nlm.nih.gov/giab/ftp/release/AshkenazimTrio/HG002_NA24385_son/v5.0q/) from the GIAB FTP site; and the HG002 nanopore R10.4.1 reads and alignments from the ONT open-data archive (s3://ont-open-data/giab_lsk114_2022.12/) under a CC BY-NC 4.0 licence. The truth files used are listed in Supplementary Table 5. [State the command used to derive the v5.0q chr1–chr22 subset and provide source data for Figs. 2–4 and every Supplementary figure with plotted values, including any access restrictions.]

Code availability

LongPhase 2 is open source under the GNU General Public License v3.0 at <https://github.com/twolinin/longphase>; the repository includes a demonstration data set and workflow. The reported results correspond to release v2.1 (commit 46ba47025b2704ab00cf54a35d3d4fead3eb148c). The GNN training code, the trained model and the scripts used for the analyses and figures are available in the same repository. [Add the training, analysis and figure-generation scripts to the public repository under an OSI-approved licence, with development-machine paths parameterized, before submission.]

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